

Phase 2

KV Cache enabled inference.py output:

```
(torch-env) [tmjoshi@e21-04 ml-systems-final-project-BaloneyGit-main]$ python inference.py
Reloaded tiktoken model from /home1/tmjoshi/.llama/checkpoints/Llama3.2-1B/tokenizer.model
#words: 128256 - BOS ID: 128000 - EOS ID: 128001
/home1/tmjoshi/ml-systems-final-project-BaloneyGit-main/ml-systems-final-project-BaloneyGit-main/inference.py:16: FutureWarning: You are using `torch.load` with `weights_only=False` (the current default value), which uses the default pickle module implicitly. It is possible to construct malicious pickle data which will execute arbitrary code during unpickling (See https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for more details). In a future release, the default value for `weights_only` will be flipped to `True`. This limits the functions that could be executed during unpickling. Arbitrary objects will no longer be allowed to be loaded via this mode unless they are explicitly allowlisted by the user via `torch.serialization.add_safe_globals`. We recommend you start setting `weights_only=True` for any use case where you don't have full control of the loaded file. Please open an issue on GitHub for any issues related to this experimental feature.
  checkpoint = torch.load(model_path, map_location="cpu")
/home1/tmjoshi/.conda/envs/torch-env/lib/python3.12/site-packages/torch/__init__.py:1144: UserWarning: torch.set_default_tensor_type() is deprecated as of PyTorch 2.1, please use torch.set_default_dtype() and torch.set_default_device() as alternatives. (Triggered internally at /opt/conda/conda-bld/pytorch_1729647329220/work/torch/csrc/tensor/python_tensor.cpp:432.)
  _C._set_default_tensor_type(t)
I believe the meaning of life is
> to live it, to taste experience to the utmost, to reach out eagerly and without fear for newer and richer experience. This is the meaning of life. To live it is to know a little of I and a little of Thou, and to love a little of both.
I am a Canadian born and raised in the

=====

Simply put, the theory of relativity states that
> 1) all observers moving with respect to each other will measure the same length of a meter stick; 2) all observers moving with respect to each other will measure the same speed of light; 3) all observers moving with respect to each other will measure the same time interval between two events. This means that the

=====

A brief message congratulating the team on the launch:

    Hi everyone,

    I just
> wanted to let you know that the new website is up and running. It's not perfect, but it's a start. We are still working on it and hope to have it all finished by the end of the year. In the meantime, please feel free to let us know what you think. We are looking forward

=====

Translate English to French:

    sea otter => loutre de mer
    peppermint => menthe poivrée
    plush girafe => girafe peluche
    cheese =>

> fromage
    oyster => oyster
    tortoise => tortue
    pineapple => ananas
    loafah => luffe
    rubber duck => canard en caoutchouc
    canary => canard
    eel => loutre
    coconut => co

=====

(torch-env) [tmjoshi@e21-04 ml-systems-final-project-BaloneyGit-main]$
```

KV Cache disabled inference.py output:

```
(torch-env) [tmjoshi@e21-04 ml-systems-final-project-BaloneyGit-main]$ python inference.py
Reloaded tiktoken model from /home1/tmjoshi/.llama/checkpoints/Llama3.2-1B/tokenizer.model
#words: 128256 - BOS ID: 128000 - EOS ID: 128001
/home1/tmjoshi/ml-systems-final-project-BaloneyGit-main/ml-systems-final-project-BaloneyGit-main/inference.py:16: FutureWarning: You are using `torch.load` with `weights_only=False` (the current default value), which uses the default pickle module implicitly. It is possible to construct malicious pickle data which will execute arbitrary code during unpickling (See https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for more details). In a future release, the default value for `weights_only` will be flipped to `True`. This limits the functions that could be executed during unpickling. Arbitrary objects will no longer be allowed to be loaded via this mode unless they are explicitly allowlisted by the user via `torch.serialization.add_safe_globals`. We recommend you start setting `weights_only=True` for any use case where you don't have full control of the loaded file. Please open an issue on GitHub for any issues related to this experimental feature.
  checkpoint = torch.load(model_path, map_location="cpu")
/home1/tmjoshi/.conda/envs/torch-env/lib/python3.12/site-packages/torch/_init__.py:1144: UserWarning: torch.set_default_tensor_type() is deprecated as of PyTorch 2.1, please use torch.set_default_dtype() and torch.set_default_device() as alternatives. (Triggered internally at /opt/conda/conda-bld/pytorch_1729647329220/work/torch/csrc/tensor/python_tensor.cpp:432.)
  _C._set_default_tensor_type(t)
I believe the meaning of life is
> to live it, to taste experience to the utmost, to reach out eagerly and without fear for newer and richer experience. This is the meaning of life. To live it is to know the meaning of life.
I believe the meaning of life is to live it, to taste experience to the utmost, to reach out eagerly

=====

Simply put, the theory of relativity states that
> 1) all observers moving with respect to each other will measure the same length of a meter stick; 2) all observers moving with respect to each other will measure the same speed of light; 3) all observers moving with respect to each other will measure the same time interval between two events. This means that the

=====

A brief message congratulating the team on the launch:

    Hi everyone,

    I just
> wanted to let you know that the new website is up and running. It's not perfect, but it's a start. We are still working on it and hope to have it all finished by the end of the year. In the meantime, please feel free to let us know what you think. We are looking forward

=====

Translate English to French:

    sea otter => loutre de mer
    peppermint => menthe poivrée
    plush girafe => girafe peluche
    cheese =>

> fromage

    oyster => oyster
    coconuts => coquilles de coco
    scuba => scuba
    arctic => arctique
    jellyfish => algue
    albatross => albatros
    lobster => crevettes
    whale => bale

=====

(torch-env) [tmjoshi@e21-04 ml-systems-final-project-BaloneyGit-main]$
```